



RV College of Engineering®
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Question Paper ID: 680233

Course Code: MA375TGR

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Mathematics of Music (Institutional Elective)

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	A concert hall has ambient noise at 40 dB. If the music reaches 90 dB, what is the signal-to-noise ratio in decibels?	02	1	2
1.2	Define achal (fixed) and chal (variable) swaras. Which swaras belong to each category and why?	02	1	1
1.3	Define just intonation. Calculate the frequency of the major third above C4 = 261.63 Hz in just intonation.	02	2	2
1.4	Define pitch class in the context of twelve-tone equal temperament. How many distinct pitch classes exist, and why are C4 and C5 considered the same pitch class?	02	2	1
1.5	Explain the hierarchical relationship: matra → akshara → avartana. If a tala has 7 aksharas and is performed in Khanda nadai, how many matras are in one avartana?	02	2	2
1.6	Adi Tala has the structure I O O with Chatusra jati. Calculate the total number of aksharas in one cycle.	02	1	2
1.7	A triangular wave has harmonics that decay as $\frac{1}{n^2}$, while a square wave decays as $\frac{1}{n}$. If both have fundamental amplitude 1, calculate the amplitude ratio of the 5th harmonic between triangle and square waves. Why does the triangle sound mellower?	02	3	2
1.8	Two sounds have the same fundamental frequency (100 Hz) but different spectral centroids (SC): Sound A has SC = 300 Hz, Sound B has SC = 800 Hz. Which sound will be perceived as brighter and why?	02	3	2
1.9	A fractal contains 5 self-similar copies when scaled by factor 3. Calculate its fractal dimension.	02	4	3
1.10	List a few factors that contribute to the perception of consonance and dissonance in musical intervals. Which of one is most directly related to psychoacoustics?	02	4	3

Part B

Question No	Question	M	CO	BT
2a	List all the swara variants in the Carnatic music system. Which swaras are fixed and which are variable?	05	1	1
2b	An acoustic sensor records a sinusoidal pressure wave given by: $p(t) = 0.08 \sin(5000\pi t)$ Taking the speed of sound as $v = 345 \text{ m/s}$, find: i) The frequency of the sound wave ii) The time period and its relation to the audible range iii) The wavelength of the wave iv) The total distance travelled by the wavefront in 0.02 seconds	06	1	2
2c	Derive the formula for the harmonic series and explain its relationship to musical timbre and consonance.	05	1	2
3a	Prove that $\log_2(3/2)$ is irrational using proof by contradiction. Explain in detail the musical consequence of it.	08	1	2
3b	Establish the semitone frequency multiplier in equal temperament tuning system. Taking $G4 = 392 \text{ Hz}$ as reference, compute the frequencies of: i) A4 ii) C5 iii) D5 iv) G5 v) E6 vi) G6	08	2	2
OR				
4a	Calculate the Pythagorean ratios and their corresponding cent measurements for the F major scale (F, G, A, Bb, C, D, E). Use F as the tonic and show all intermediate steps.	08	2	2
4b	Obtain the total number of possible Thaats systems in Hindustani music if all combinatorial possibilities were considered (similar to Melakartha derivation). Explain why only 10 Thaats are used in practice.	08	1	2
5a	Explain the concept of gati and its relationship to rhythmic subdivision in Carnatic music. How does changing gati mathematically alter the density of notes within a fixed tala structure?	08	1	1
5b	Construct three tihais for 2 cycles of Misra Chapu Tala (7 aksharas per cycle) in Khanda nadai. Apply the formula $3P + 2G = T$ to find valid P and G combinations. Show sample rhythmic patterns for each.	08	2	2
OR				
6a	A tala has structure I U O (Jhampa Tala) with Tisra jati. Calculate the number of aksharas in one cycle. If this tala is performed at three different tempos (vilambit = 50 bpm, madhya = 100 bpm, drut = 150 bpm) in Chatusra nadai, calculate: i) total matras per cycle, ii) duration of one avartana in seconds at each tempo, iii) duration of one akshara at each tempo. Present results in tabular form. Note: BPM (beats per minute) refers to aksharas per minute, i.e., one beat = one akshara.	08	1	1
6b	A korvai has purvanga of 14 matras and uttaranga of 7 matras. i) If repeated 3 times (standard korvai structure), calculate the total length in matras. ii) Does it fit exactly in 2 cycles of Misra Chapu Tala (7 aksharas per cycle, Chatusra nadai)? Justify your answer.	08	2	2

- iii) If it doesn't fit exactly, calculate the shortfall or excess in matras.
- iv) Alternatively, determine how many cycles of Misra Chapu would be needed to accommodate this korvai exactly.

7a	Using the nonlinear distortion equation $y(t) = a_1x(t) + a_2x^2(t)$ and the trigonometric identity $\cos(A)\cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$, derive the mathematical explanation for summation tones. If two pure tones at frequencies f_1 and f_2 are input, show that the output contains the summation frequency $f_1 + f_2$.	08	3	3
7b	When two notes at 440 Hz (A4) and 660 Hz (E5) are played simultaneously: (i) Calculate the first-order difference tone. (ii) Calculate the first-order summation tone. (iii) Identify the musical interval represented by 440 Hz and 660 Hz. Also, explain why combination tones are audible even at moderate volumes in certain acoustic spaces like cathedral organs.	08	3	2
	For a square wave $f(t) = \begin{cases} k, & 0 < t < \frac{T}{2}, \\ -k, & \frac{T}{2} < t < T \end{cases}$, derive its Fourier series representation showing that it contains only odd harmonics with amplitudes decreasing as $1/n$. Explain why the square wave sounds different from a sawtooth wave despite both being periodic.	08	3	2

OR

8a	Explain the following sound synthesis methods: i) Additive Synthesis - principle, building complex timbres, instrument example ii) Subtractive Synthesis - method, filter types (low-pass, high-pass, band-pass), popularity in analog iii) FM Synthesis - carrier/modulator concept, modulation index effects, example sounds iv) Granular Synthesis - grain construction, parameters (size, density, pitch), characteristic effects	08	3	3
8b	For a pulse wave (rectangular wave with duty cycle $d = 0.25$) $f(t) = \begin{cases} 4, & 0 < t < \frac{T}{4} \\ 0, & \frac{T}{4} < t < T \end{cases}$ derive its Fourier series representation. Show that it contains both odd and even harmonics. Explain how this narrow pulse (duty cycle 0.25) produces a different timbre compared to a square wave (duty cycle 0.5).	08	3	2
9a	(i) Define critical bandwidth using Zwicker's formula. Calculate the critical bandwidth at 2000 Hz. (ii) Using your result from (i), analyze whether an equal temperament major third interval (2000 Hz and _____ Hz) would be perceived as consonant or dissonant. Explain the relationship between critical bands and the perception of roughness.	08	4	3
9b	i) Define a first-order Markov chain and explain the Markov property in the context of musical melody generation. ii) Construct a valid 3×3 transition matrix P for generating melodies using notes {C, D, E} with the following rules: • Staying on the same note: probability 0.3 for all notes • Stepwise motion: total probability 0.5, distributed equally among all available stepwise neighbors • Skip motion: total probability 0.2, distributed equally among all available skip destinations	08	4	4

- If a transition type is impossible, redistribute that probability equally among available stepwise neighbors

Verify your matrix is row-stochastic.

iii) Given the initial state vector $\pi_0 = [0.5, 0.3, 0.2]$ (starting probabilities for C, D, E), calculate the probability distribution after one step ($\pi_1 = \pi_0 P$).

iv) Starting from note D, what is the probability of generating the specific melody sequence $D \rightarrow E \rightarrow D \rightarrow C$? Show your calculation using the transition matrix.

OR

i) Explain the missing fundamental phenomenon in psychoacoustics. Describe the physiological and cognitive mechanisms that allow humans to perceive a pitch that is not physically present in the sound.

ii) A complex tone contains harmonics at 300 Hz, 450 Hz, 600 Hz, and 750 Hz but lacks the fundamental. What pitch will a listener perceive and why? Show how the brain reconstructs the missing fundamental from the harmonic series.

10a

08 4 3

iii) If the 300 Hz component is removed, leaving only 450 Hz, 600 Hz, and 750 Hz, what pitch will be perceived? Explain how the brain still identifies the fundamental even with incomplete harmonic information.

iv) Discuss practical applications of the missing fundamental phenomenon in telephone systems & audio compression, small speaker design, and bass enhancement technologies. Why is this phenomenon crucial for modern audio engineering?

10b

08 4 4

A four-note melody system has the following note probabilities: Sa (0.4), Ri (0.3), Ga (0.2), Ma (0.1). i) Calculate the Shannon entropy H in bits per note. ii) Calculate the maximum entropy (H_{max}) and redundancy (R) for this system. iii) If a second melody system has probabilities Sa (0.25), Ri (0.25), Ga (0.25), Ma (0.25), calculate its entropy and redundancy. Compare both systems in terms of predictability and listener engagement. iv) A composer wants to create a more surprising melody by adjusting the probabilities. Suggest new probabilities that would increase entropy to at least 1.9 bits, and verify your answer through calculation.